

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Defined and initiated the **AI strategy** for core Bitcoin mining platforms (MOS & MDK), focusing on **operational efficiency** and measurable site-level outcomes.
- Designed a **standardized data baseline and schema** for mining-site telemetry and events, ensuring structured, normalized, and accessible data for AI/ML use cases.
- Developed and deployed **ML models** for **predictive failure analysis** on mining hardware, reducing unscheduled downtime by an average of **18%** across monitored sites.
- Engineered **Python data pipelines** on AWS Lambda and S3 to process high-volume telemetry, replacing a brittle scheduled-batch job and handling approximately **3,500 events/sec** at peak hours.
- Collaborated with site-operations teams to validate model performance and gather feedback on **energy-efficiency insights**, leading to a **7%** reduction in power consumption for specific mining rigs.
- Integrated **ML-driven optimization recommendations** into the MOS platform via REST APIs, enabling automated adjustments to mining parameters that improved hash rate stability by **~5%**.

SKILLS

AI/ML & Optimization – AI Strategy, Machine Learning, Mathematical Optimization, Predictive Maintenance, Anomaly Detection, Performance Optimization, Energy Efficiency

Data Engineering & Architecture – Data Baseline Design, Data Schema Design, Data Modelling, Data Pipelines, ETL, SQL (PostgreSQL, MySQL, Oracle), AWS (S3, Lambda)

Programming & Backend – Python, Java, Spring Boot, REST APIs, Flask, FastAPI

Cloud & Infrastructure – AWS (EC2, CloudWatch), Docker, Linux

MLOps & Deployment – Model Deployment, Model Evaluation, CI/CD (Jenkins), Git

Tools & Methodologies – Jira, Confluence, Agile/Scrum

PROJECTS

Mining Operations AI-Driven Optimization Engine

- Designed and implemented a Python-based system to ingest and analyze real-time sensor data from mining hardware, focusing on predicting component failures.
- Trained a **LightGBM classifier** on a dataset of **~450,000 operational logs** to identify patterns preceding hardware malfunctions, achieving an **AUC of 0.92**.
- Developed a REST API using FastAPI to serve failure predictions, integrating with the existing site monitoring dashboard and triggering alerts for maintenance teams.

Telemetry Data Standardization for Mining Sites

- Architected a standardized data schema in PostgreSQL for collecting diverse telemetry streams (temperature, hash rate, power consumption, error codes) from thousands of mining units.
- Built Python scripts to transform and load data from various sources into the standardized schema, ensuring data quality and consistency for downstream analytics and ML workloads.
- Collaborated with operations analysts to define key metrics and reporting requirements, enabling the creation of **15+ new operational performance dashboards**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)